STAT8480 Project Guidelines

Fall 2026

This document provides instructions on writing your project proposal and general guidelines for carrying out a data mining project.

Project Proposal:

Your project proposal should be approximately 1 page long, single spaced. The purpose of the proposal is to make sure that you are on the right track and to give me enough information so that I can give you useful feedback.

In your proposal you should cover the following items:

- Preliminary title and list of students working on the project (1-3 students per project; 4 or more with special permission).
- Brief description of what you plan to do.
  - What problem are you trying to solve?
  - How do you formulate the problem as a data mining problem? What exactly are you trying to predict and how will you evaluate your results? How will you know if your results are good? What can you compare them to? It is critical that your problem is **well-defined**.
  - What data sets do you plan to use? If you must do significant work to get the data or convert it into the proper format, then describe the process and approximate effort required. How many observations are in the data set? How many variables?
  - If sufficiently valuable, list a few related research papers.

Proposal Submission

The project proposal is due from groups by **11:59 PM on Saturday Oct. 17** via Brightspace.

Project Procedure:

In general, a data mining project is carried out in two phases with each phase described below:

Phase 1: Data Audit and Descriptive Analysis

Data Audit:

Your primary task in this phase is to try to understand the nature of the data and relationship among the variables. The more time you spend at this phase, the better your models will be in the next stage. You are encouraged to examine each variable individually and determine its nature

and strength of relationship with the target variable. While statistical significance of any relationship is relevant, the actual strength of relationship and whether the relationship makes sense or interesting/surprising (based on your understanding of the project problems) are more important in this phase of analysis.

Descriptive Analysis:

As you start on this phase, first spend some time reading about background information, the variable descriptions in the data set and do some thinking about these variables and how they may relate to the target variable.

Then you should first look at the frequency distribution for each categorical independent (input) variable and see if there are any inconsistencies or errors in the data. You should also try to get a sense of any rare or extreme values in the data. Next, it is important to identify and distinguish ordinal and nominal variables among the categorical inputs. Furthermore, you may consider to look at the distribution of the target variable across all the categories of a categorical input variable. If there is any variation in the target distribution that shows interesting (and hopefully meaningful) pattern, the related categorical input variables have great potential to be valuable predictors for models. For a categorical input with a large number of categories, you can consider to consolidate the categories that exhibit similar patterns in relationship to the target variable.

For interval-level input variables, you might consider reporting the average values (and/or other related statistics) of the input variables grouped separately by the levels of the target variable. Again, if any of these patterns of means on input variables shows interesting (and hopefully meaningful) pattern, it is a good signal for valuable predictors. For interval inputs with extreme distributions, the appropriate transformations should be investigated and applied.

For any input variable with missing values, you should consider to use appropriate treatment including imputation methods, as needed.

In addition, explore the variable importance with different measures and prepare it as a guide to filtering which may be used as inputs in Phase II.

Phase I Submission

Phase 1 complete report is due from groups by **11:59 PM on Saturday Nov. 14** via Brightspace. The report should provide a detailed introduction to your project problems and goals, and a complete description of the data you chose and the data exploration and preparation you did in Phase I. Please include all supporting evidence (such as plots, summary statistics, exploratory models, etc.) properly labeled and annotated in the report.

In addition, each team member must individually submit a brief report (approximately one paragraph) summarizing the team's roles and an evaluation of individual contributions to the Phase I deliverables.

All teams will give a Phase I presentation on your projects in class before the Phase I report is due. The presentation guideline will be revealed later.

Phase 2: Predictive Modeling

Your task is to develop model(s) that will help us predict fairly accurately the target variable. Your understanding of data in Phase 1 is expected to play a critical role as you develop models in Phase 2. Here are some things to consider:

- Balance (oversample) as appropriate.
- Split data into training-validation-test sample.
- Recode categorical variables with large number of categories to a smaller number of meaningful categories.
- Transform variables and impute missing values as needed.
- Create interaction or other derived variables (as appropriate).
- Run different predictive models on training, validation, test data
  - Logistic Regression
  - Decision Trees
  - Random Forest
  - Others as appropriate
- Identify the best performing predictive model
  - Choose appropriate model assessment metrics
  - Report model performance across training-validation-test data
  - Explain the best model
    - Translate your findings to meaningful rules

Phase II Submission

Final Report: (All the final report deliverables are due on Brightspace by **11:59pm, Friday Dec. 18**)
The final report should contain a comprehensive account of your project. It should be thorough, yet concise.

(i) Basic Requirements:

1. Your report should be between 5-7 pages long (not including appendix).
2. Font size 11-12, 1″ margins, 1-1.5 line spacing.
3. Your report should be submitted in MS Word .docx or .pdf.

(ii) Sections to include: The paper need not be organized exactly as described below, but it should be quite similar, since the outline below is standard for a conference paper.

1. Introduction

This should provide the reader with the necessary background information about why your project is an interesting and worthwhile project, and where the project fits into real life. Give a clear project statement that explains the goals of the project.

2. Methodology

Explain in depth the model(s) you are using to solve the problem, including the model architecture and hyperparameters. List out all variables and their data types, which is probably best done with a table. Describe the data you chose, the data preparation you did, evaluation metrics, data mining algorithms explored, and any other related details in Phase I and II of the project. There will usually be a subsection for each of the sub-topics just mentioned. Please include any graphics, plots, charts that you find helpful in demonstrating the topics above.

3. Results Present the model results and a discussion and analysis of the results.

4. Conclusion

Bring together all you have learned and summarize the main takeaways of your project, especially the real-life applications. Discuss limitations and avenues for future work.

(iii) Submission

- ❖ Project zip file: Please organize your final model R script(s) (or R markdown, Quarto) and the data source in an RStudio project. Save your final report in the same project directory and submit the zip file of the project directory to Brightspace.
- ❖ Teamwork report: This is the only individual portion of the final report. Each person in the team must write their own short report (about a paragraph) describing the role of each team member and how well, in your opinion, they contributed to the final product. These reports should be submitted individually to Brightspace.

Presentation:

All teams will give a final presentation on your projects to the class at the end of the semester. The presentation guideline will be revealed later.

More tips for conducting a successful project:

- **Keep a journal/notebook of your progress.** Every time your group meets, keep notes of what was decided upon, who is going to do what for the next meeting, and so on. Also remind yourself of everything you did at the end of each day when you work on the project. Summarize what you accomplished, what choices you made regarding your model, what sources you read, what questions you have, and what you plan to do next.
- Remember you are never quite finished with a given step in the modeling process. Attempt to improve models using new data preparation approaches, trying multiple modeling techniques and being creative. **Repeat the above** on your new version of the model as you go along.
- Remember I am available for help. We will have set meetings in class and outside of class but *please come and talk to me at any point in the process*.

Important Dates:

Project proposal: due Saturday, Oct. 17, 11:59pm.
Phase I report: due Saturday, Nov. 14, 11:59pm.
Final report: due Friday, Dec. 18, 11:59pm.